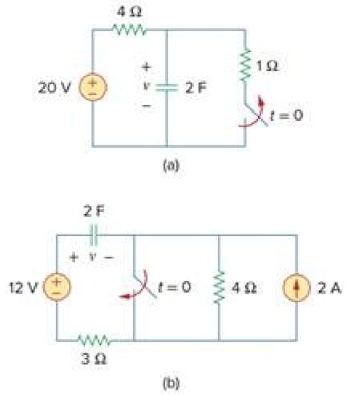
Chapter 7, Problem 39P

Problem

Calculate the capacitor voltage for $t<0$ and $t > 0$ for each of the circuits in Fig. 7.106.

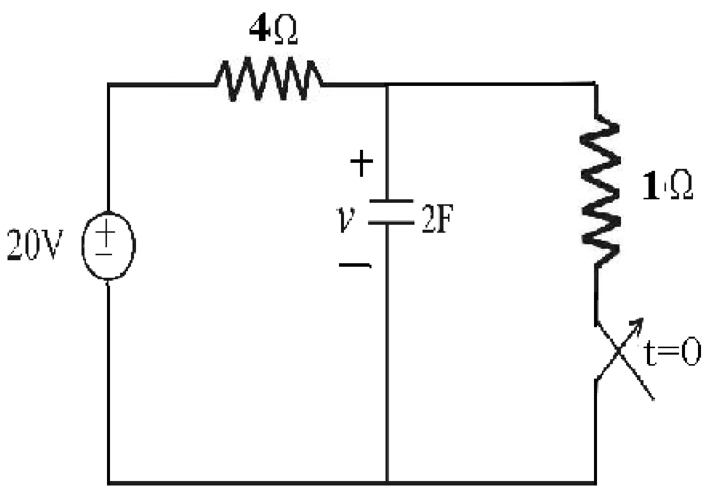
Fig. 7.106:



Step-by-step solution

Step 1 of 7

(a)



Step 2 of 7

(i)

For t < 0, the circuit is in steady state condition, so capacitor is replaced by open circuit

So,

$$v_C(0) = 20\times\left(\frac{1}{1+4}\right)$$

$$v_C(0) = 20\times\frac{1}{5}$$

$$\boxed{v_C(0) = 4\text{V}}$$

Step 3 of 7

(ii)

For t > 0, we have

$$v_C(t) = A - (A-B)e^{\frac{-t}{\tau}}$$

Where,

$$A = v_C(t \to \infty) \quad [\because C \text{ is open}]$$

$$\therefore A = 20 \text{ V}$$

$$B = v_C(t \to 0^+)$$

$$= v_C(t \to 0^-)$$

$$\therefore B = 4 \text{ V}$$

Step 4 of 7

Time constant,

$$\tau = R_{eq}\, C$$

$$\tau = (4)\times(2)$$

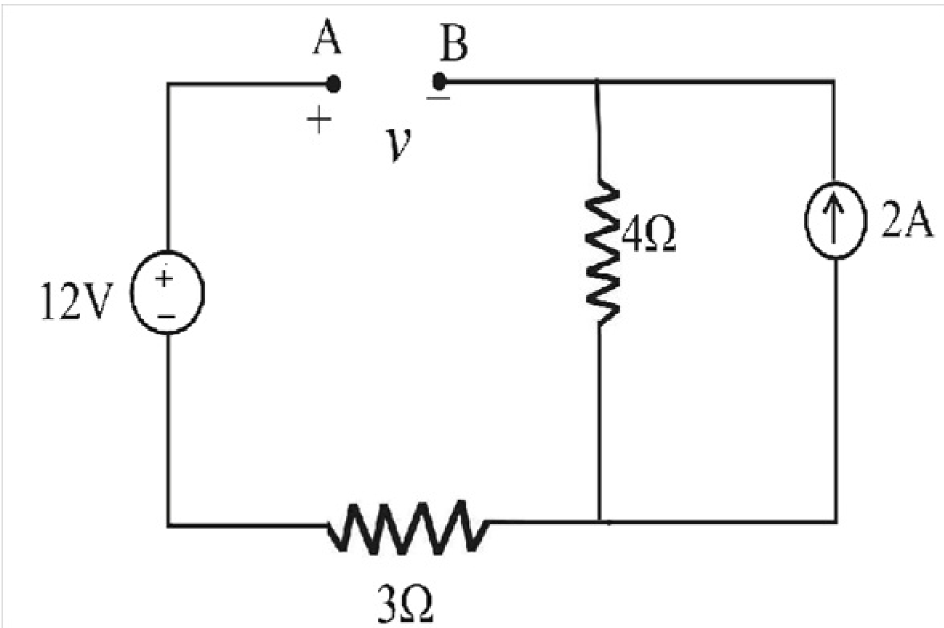
$$\therefore \tau = 8\text{s}$$

So, $v_C(t) = 20 - (20-4)e^{\frac{-t}{8}}$ V

$$\therefore \boxed{v_C(t) = 20 - 16\, e^{\frac{-t}{8}} \text{ V}}$$

Step 5 of 7

(b)



Step 6 of 7

(i)

For t < 0, steady state condition the capacitor is open circuited,

$V_C(0^-)$ is given by

$v_A = 12V$

$v_B = 2 \times 4$

$\therefore v_B = 8 \text{ V}$

So,

$v_C = v_A - v_B$

$\therefore \boxed{v_C(0) = 4 \text{ V}}$

(ii)

For t > 0 abrupt change ; so response is,

$v_C(t) = A - (A - B)e^{\frac{-t}{\tau}}$

$A = v_C(t \to \infty)$

$A = 12 \text{ V}$

$B = v_C(t \to 0^+)$

$v_C(t \to 0^-) = 4 \text{ V}$

$\therefore B = 4 \text{ V}$

Step 7 of 7

Time constant,

$\tau = R_{eq}C$

$\tau = 3 \times 2$

$\therefore \tau = 6 \text{ s}$

$\therefore \boxed{v_C(t) = 12 - 8\, e^{\frac{-t}{6}} \text{ V}}$